

DSA 8020 R Lab 5: Analysis of covariance and Non-linear Regression

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Analysis of covariance: Salaries for Professors

The 2008-09 nine-month academic salary for Assistant Professors, Associate Professors, and Professors in a college in the U.S. was collected as part of the ongoing effort of the college’s administration to monitor salary differences between male and female faculty members.

Load the dataset

Code:

```
library(carData)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

data(Salaries)
head(Salaries)

##      rank discipline yrs.since.phd yrs.service  sex salary
## 1    Prof          B           19          18 Male 139750
## 2    Prof          B           20          16 Male 173200
## 3 AsstProf          B            4            3 Male  79750
## 4    Prof          B           45          39 Male 115000
## 5    Prof          B           40          41 Male 141500
## 6 AssocProf        B            6            6 Male  97000
```

Description of the variables

- **rank:** a factor with levels Assistant Professor (“AsstProf”); Associate Professor (“AssocProf”); Full Professor (“Prof”)

- discipline: a factor with levels A (“theoretical” departments) or B (“applied” departments)
- yrs.since.phd: years since her/his PhD
- sex: a factor with levels “Female” and “Male”
- salary: nine-month salary, in dollars

Exploratory Data Analysis

1. Identify the numerical variables and categorical variables in this data set

Answer:

Numerical Vars

yrs.since.phd, salary

Categorical Vars

rank, discipline, sex

2. Summarize each variable numerically and graphically, briefly describe your findings

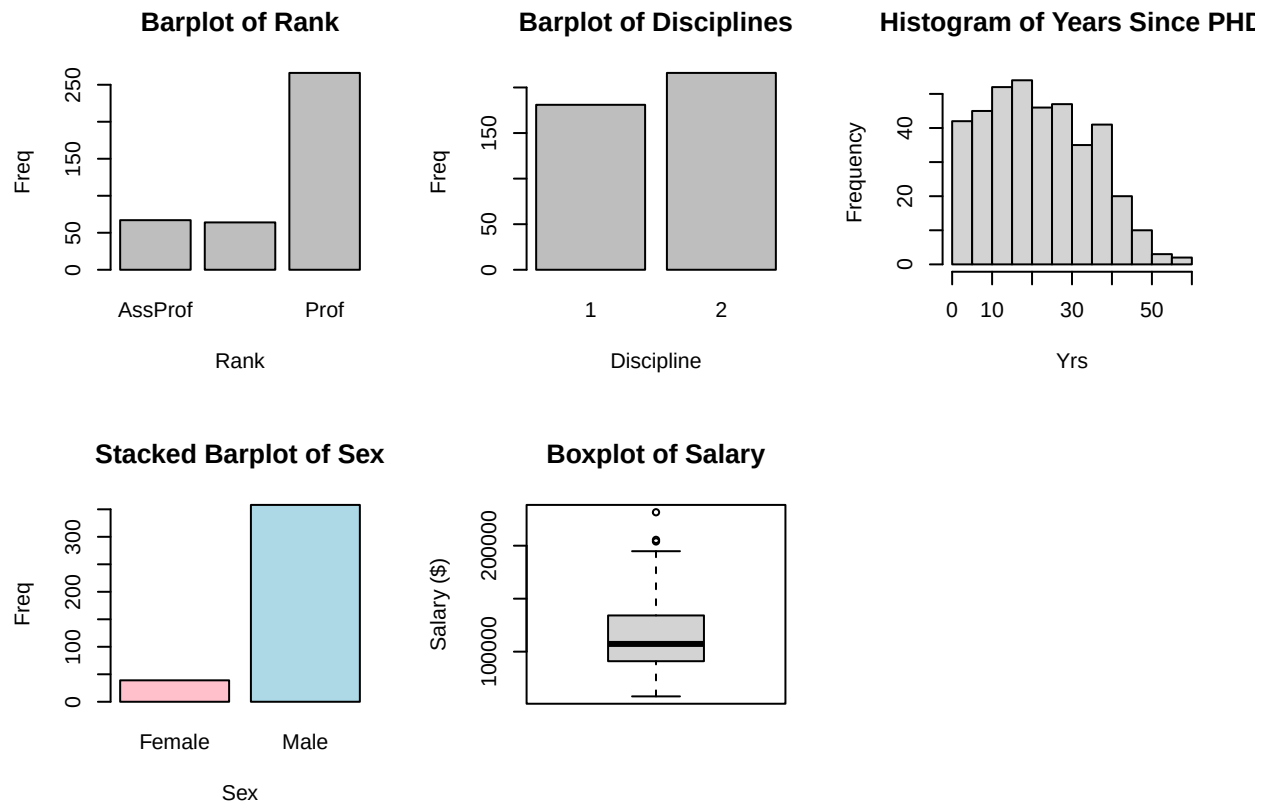
Code:

```
vars <- c("rank", "discipline", "yrs.since.phd", "sex", "salary")
data <- Salaries[, vars]
data <- data %>%
  mutate(rank = as.numeric(rank),
         discipline = as.numeric(discipline),
         sex = as.numeric(sex))

# numerical summary
summary(data)
```

```
##      rank      discipline yrs.since.phd      sex
## Min.   :1.000   Min.   :1.000   Min.    : 1.00   Min.   :1.000
## 1st Qu.:2.000   1st Qu.:1.000   1st Qu.:12.00   1st Qu.:2.000
## Median :3.000   Median :2.000   Median :21.00   Median :2.000
## Mean   :2.501   Mean   :1.544   Mean   :22.31   Mean   :1.902
## 3rd Qu.:3.000   3rd Qu.:2.000   3rd Qu.:32.00   3rd Qu.:2.000
## Max.   :3.000   Max.   :2.000   Max.   :56.00   Max.   :2.000
##
##      salary
## Min.   : 57800
## 1st Qu.: 91000
## Median :107300
## Mean   :113706
## 3rd Qu.:134185
## Max.   :231545
```

```
# graphing summary
par(mfrow=c(2, 3))
barplot(table(data$rank), main = "Barplot of Rank", xlab = "Rank", ylab = "Freq", names.arg = c("AssPro", "Dis", "Yrs", "Sex", "Salary"))
barplot(table(data$discipline), main = "Barplot of Disciplines", xlab = "Discipline", ylab = "Freq")
hist(data$yrs.since.phd, main = "Histogram of Years Since PHD", xlab = "Yrs")
barplot(table(data$sex), main = "Stacked Barplot of Sex", xlab = "Sex", ylab = "Freq", names.arg = c("F", "M"))
boxplot(data$salary, main = "Boxplot of Salary", ylab = "Salary ($)")
```



Answer:

Rank

There are more full professors than assistant or associate professors.

Discipline

Both A and B seem mostly balanced, with B having slightly greater occurrences.

yrs.since.phd

Most staff have between 10 and 40 years of elapsed time since their PhD. Distribution has slight right skew, with very few staff having 50 or more years since their PhD.

Sex

Highly imbalanced, with Males greatly dominating the staff roster.

Salary

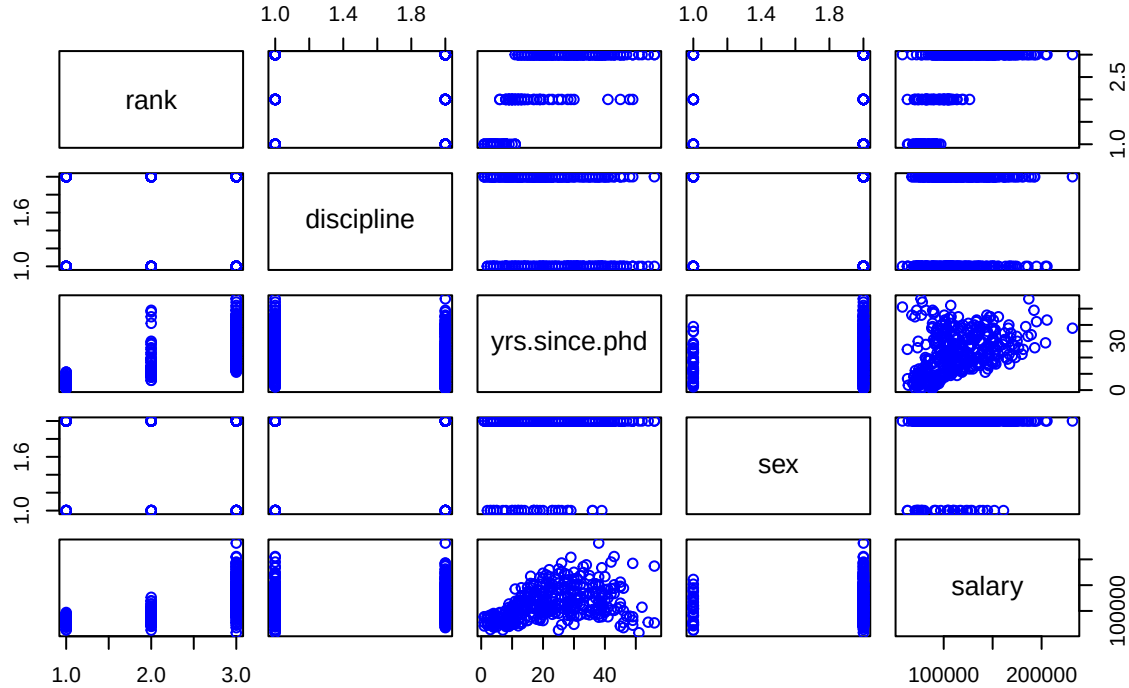
Salaries range from \$57k to \$231k, with a median of \$107k and an average salary of \$113k.

3. Create a scatterplot matrix and briefly describe your findings

Code:

```
pairs(data, main = "Scatterplot", col = "blue")
```

Scatterplot



Answer:

- rank, discipline, sex
These categorical vars are not well represented in a scatterplot.
- yrs.since.phd vs salary
Clear positive correlation, higher salary with more years since phd.
- rank vs salary
Slightly linear association indicating higher rank correlates with higher salary.
- sex vs salary
ANOVA must be used to determine if there is significant difference in salaries between genders.

Model Fitting

4. Fit a multiple linear regression model (MLR) with `yrs.since.phd`, `discipline`, `rank`, and `sex` as predictors. Write down the fitted regression equations for each category (e.g., Female, Assistant Professor, theoretical departments). There are 12 categories in total

Code:

```
model <- lm(salary ~ yrs.since.phd + discipline + rank + sex, data = Salaries)
summary(model)
```

```
##
## Call:
## lm(formula = salary ~ yrs.since.phd + discipline + rank + sex,
##     data = Salaries)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -67451 -13860  -1549   10716   97023
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  67884.32    4536.89   14.963 < 2e-16 ***
## yrs.since.phd    61.01     127.01    0.480  0.63124
## disciplineB   13937.47    2346.53    5.940 6.32e-09 ***
## rankAssocProf 13104.15    4167.31    3.145  0.00179 **
## rankProf      46032.55    4240.12   10.856 < 2e-16 ***
## sexMale        4349.37     3875.39    1.122  0.26242
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22660 on 391 degrees of freedom
## Multiple R-squared:  0.4472, Adjusted R-squared:  0.4401
## F-statistic: 63.27 on 5 and 391 DF,  p-value: < 2.2e-16
```

Answer:

- Fitted equation for salary

$\hat{\text{salary}} = 67884.32 + 61.01 * (\text{yrs.since.phd}) + 13937.47 * (\text{disciplineB}) + 13104.15 * (\text{AssocProf}) + 46032.55 * (\text{Prof}) + 4349.37 * (\text{Male})$

1. **Female, Assistant, Theoretical** $\hat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd})$
2. **Female, Assistant, Applied** $\hat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 13937.47 \times (\text{disciplineB})$
3. **Female, Associate, Theoretical** $\hat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 13104.15 \times (\text{AssocProf})$
4. **Female, Associate, Applied** $\hat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 13104.15 \times (\text{AssocProf}) + 13937.47 \times (\text{disciplineB})$
5. **Female, Full, Theoretical** $\hat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 46032.55 \times (\text{Prof})$
6. **Female, Full, Applied** $\hat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 46032.55 \times (\text{Prof}) + 13937.47 \times (\text{disciplineB})$
7. **Male, Assistant, Theoretical** $\hat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 4349.37 \times (\text{Male})$
8. **Male, Assistant, Applied** $\hat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 13937.47 \times (\text{disciplineB}) + 4349.37 \times (\text{Male})$
9. **Male, Associate, Theoretical** $\hat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 13104.15 \times (\text{AssocProf}) + 4349.37 \times (\text{Male})$
10. **Male, Associate, Applied** $\hat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 13104.15 \times (\text{AssocProf}) + 13937.47 \times (\text{disciplineB}) + 4349.37 \times (\text{Male})$
11. **Male, Full, Theoretical** $\hat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 46032.55 \times (\text{Prof}) + 4349.37 \times (\text{Male})$
12. **Male, Full, Applied** $\hat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 46032.55 \times (\text{Prof}) + 13937.47 \times (\text{disciplineB}) + 4349.37 \times (\text{Male})$

-
5. State the model assumptions in the previous regression model

Answer:

6. Now fit another MLR with `yrs.since.phd`, `discipline`, `sex` and their interactions. Write down the fitted regression equations for each category

Code:

Answer:

Non-linear Regression: An Simulated Example

Suppose the response y depends on the predictor t in the following form:

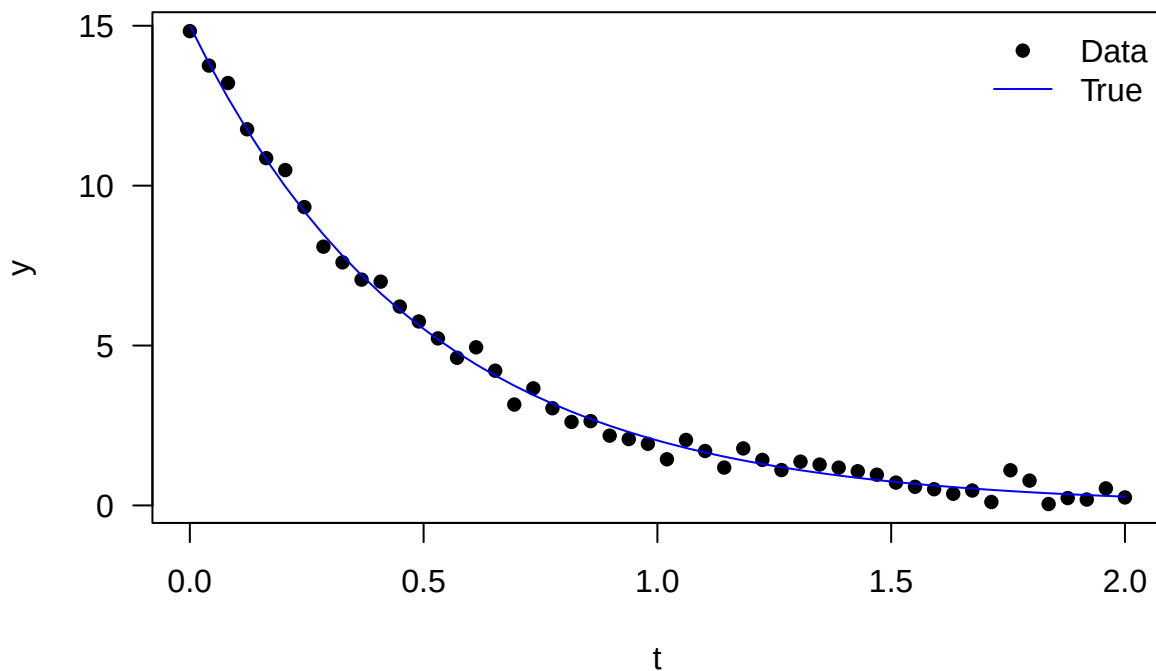
$$y = \alpha \exp(-\beta t) + \epsilon,$$

where $\epsilon \sim N(0, \sigma^2)$, and the true α , β , and σ^2 are 15, 2 and 0.16, respectively. First, let's simulate some data points from this nonlinear model:

Code:

```
alpha = 15; beta = 2; sigma.sq = 0.09
n <- 50
t <- seq(0, 2, len = 50)
set.seed(123)
y <- alpha * exp(-beta * t) + rnorm(n, sd = sqrt(sigma.sq))
data <- data.frame(y = y, t = t)

plot(t, y, las = 1, pch = 16)
lines(t, alpha * exp(-beta * t), type = "l", col = "blue")
legend("topright", legend = c("Data", "True"), pch = c(16, NA), lty = c(NA, 1),
      col = c("black", "blue"), bty = "n")
```



7. Use the `nls` function to obtain nonlinear least-squares estimates $\hat{\alpha}$, $\hat{\beta}$, and $\hat{\sigma}^2$. To use `nls`, provide `formula = y ~ alpha * exp(-beta * t)`, `start = list(alpha = alpha_0, beta = beta_0)`, where `alpha_0` and `beta_0` are initial guesses of the parameters α and β

Code:

Answer:

8. Write down the fitted equation and the estimated variance $\hat{\sigma}^2$

Answer:

9. Apply the natural log transformation to the simulated response, then fit a simple linear regression.
Back-transform to obtain the fit on the original scale

Code:

Answer:

10. Comparing the nonlinear regression method and the linear regression with log-transformed response, which method would you prefer in this example? Explain your answer

Answer: